

LoRaWAN

LoRaWAN is an abbreviation for Low Power, Wide Area Networking protocol. A Wireless Spread Spectrum Modulation Technology, which is designed for Low Power and Wide Area Network. It operates in an unlicensed spectrum while ensuring Global Interoperability and High Scalability, along with a multitude of other handy functionalities.



Scalable
Network



Dual-Layered
Security



Reduced Cost
of Deployment

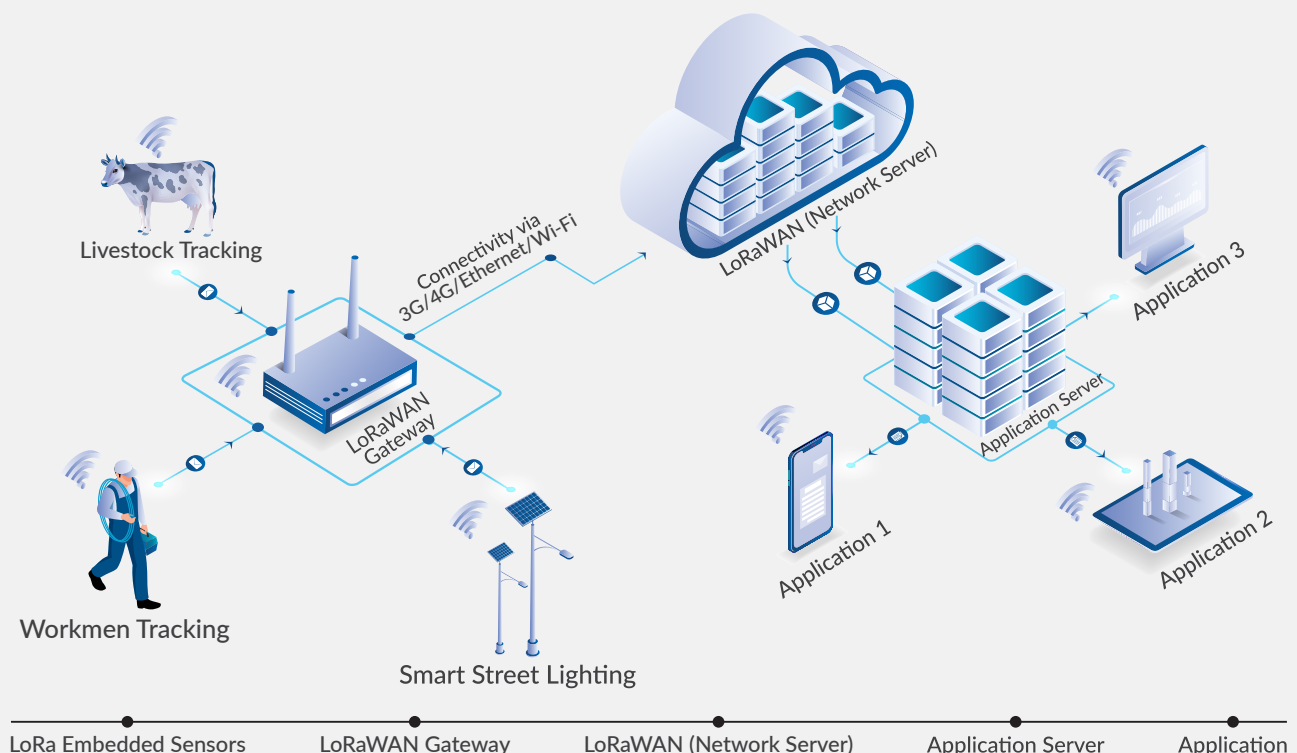


Standardized
Solution for
IoT Applications

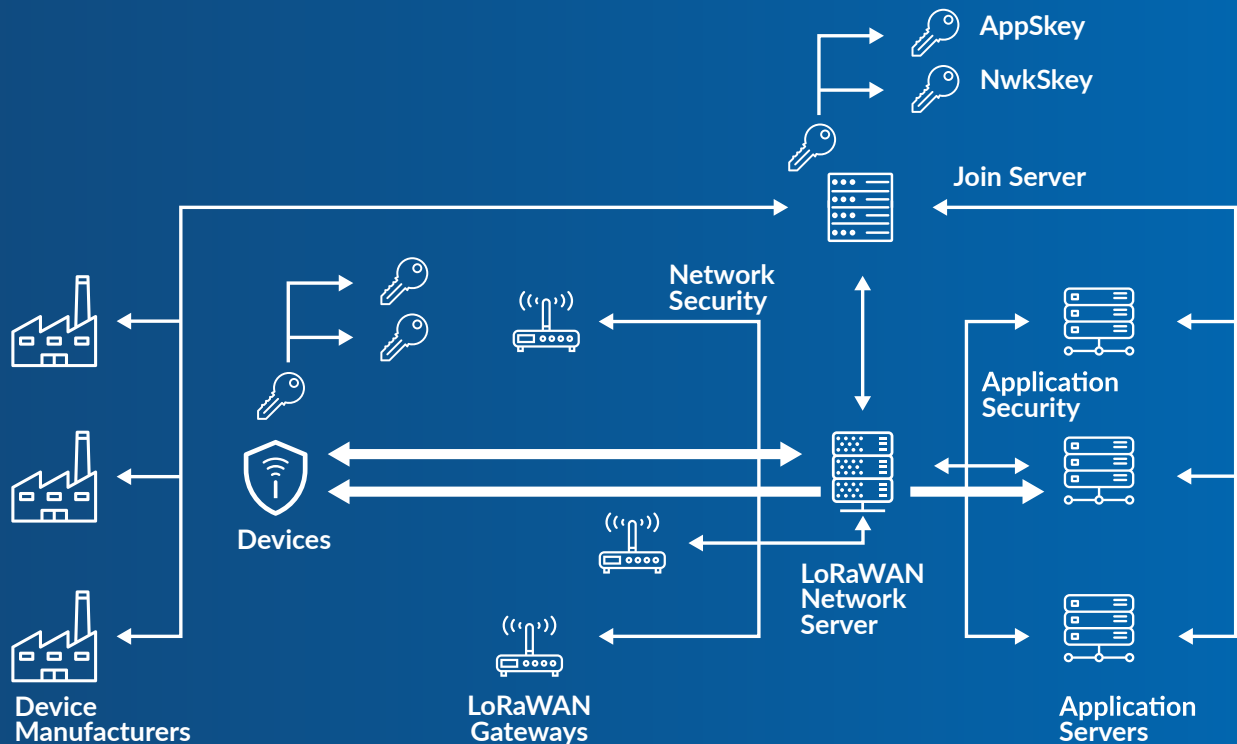
Advanced Features (LoRaWAN Server)

- LoRaWAN Versions (1.0x and 1.1x) Supported
- Supported Device Classes (Class A, B, and C)
- Remote Multicasting
- Fragmented Data Block Transfer supported
- Backed by Data Log and Live Frames
- Firmware Over-the-Air Updates
- Multiple Data Outputs for End Applications
- Third-Party Output Format Integration
- Provides Geolocation
- Public Cloud, Private Cloud, and on-premise deployment options
- Secured Communication with AES 128 and SSL/TLS

The following diagram shows a systematic mechanism on how a LoRaWAN ecosystem works.



The following diagram shows how a LoRaWAN Security Apparatus works.



Use Cases



Smart Cities

- Smart Parking
- Smart Waste Management
- Smart Street Lighting



Industrial IoT

- Process Control and Monitoring
- Smart Manufacturing
- Predictive Maintenance



Utilities

- Smart Metering
- Smart Grid Management



Agritech

- Livestock Tracking
- Livestock Monitoring



Mining

- Conveyor Belt Applications
- Emergency Underground Communication



Warehousing

- Data Feedback from any location
- Real-time Fleet Monitoring



Logistics

- Fleet Management
- Asset Management and Tracking



Environmental Monitoring

- Flood Monitoring/Alerts (Disaster Prevention & Management)
- Water Pollution, Air Pollution, & Noise Pollution



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